PHY101: Introduction to Physics-1

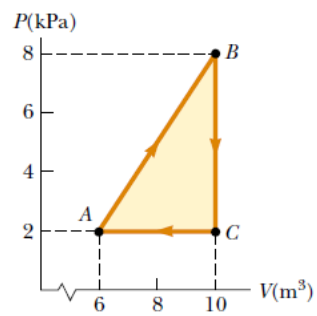
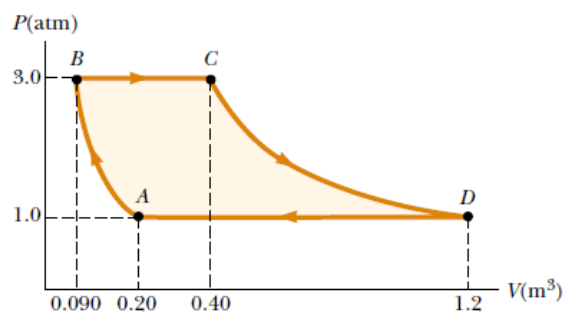
Monsoon, 2018

Tutorial-14

1. A thermally insulated vessel containing a gas whose molar mass is equal to M and the ratio of specific heats γ moves with a velocity v. Find the gas temperature increment resulting from the sudden stoppage of the vessel.

2. A gas consists of molecules of mass m and is at a temperature T. Making use of the Maxwell velocity distribution function, find the corresponding distribution of the molecules over the kinetic energies $\mathcal{E}$. Determine the most probable value of the kinetic energy $\mathcal{E}p$. Does $\mathcal{E}p$ correspond to the most probable velocity?

3. A gas is taken through the cyclic process described in the figure. (a) Find the net energy transferred to the system by heat during one complete cycle. (b) If the cycle is reversed, that is, the process follows the path *ACBA*, what is the net energy input per cycle by heat?



4. A sample of an ideal gas goes through the process shown in figure. From *A* to *B*, the process is adiabatic; from *B* to *C*, it is isobaric with 100 kJ of energy entering the system by heat. From *C* to *D*, the process is isothermal; from *D* to *A*, it is isobaric with 150 kJ of energy leaving the system by heat. Determine the difference in internal energy $E_{int,B} - E_{int,A}$.